NATIONAL PRACTICAL EXAMINATION FOR TECHNICAL SECONDARY SCHOOLS (TSS)

SCHOOL YEAR 2024-2025

SUMMATIVE/INTEGRATED ASSESSMENT

SECTOR: ICT AND MULTIMEDIA

TRADE: SOFTWARE DEVELOPMENT

E.S. SAINT JOSEPH LE TRAVAILLEUR

Integrated Assessment Project: Develop a web application using Node JS, React JS and MySQL/MongoDB

COMPETENCES ASSESSED

S/No	Code	Competence names	RQF Level
1	SWDPR301	Analyze Project Requirements	Level 3
2	SWDUX301	Design UI/UX	Level 3
3	SWDJF301	Apply JavaScript	Level 3
4	SWDDA401	Apply Data Structure and Algorithm Fundamentals Using JavaScript	Level 4
5	SWDBS401	Design Backend System	Level 4
6	SWDDD401	Develop Database	Level 4
7	SWDBD401	Develop a Backend Application using Node Js	Level 4
8	SWDFA501	Develop Frontend Application using React.JS	Level 5
9	GENQA501	Apply Quality Assurance	Level 5
10	SWDND501	Develop NoSQL Database	Level 5

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1. INTEGRATED SITUATION:

Scenario

SmartPark is company located in Rubavu District, western province of Rwanda. It provides multiple car-related services. It is struggling with inefficiencies due to its manual, paper based system for managing parking space sales. The parking space manager records slot details such as numbers and status, and car details such as plate number, entry time, exit time, driver's name and phone number on paper. Upon a car's exit the manager manually calculates and records the parking duration, determine the bill, and update the slot status for new arrivals. This process is slow, prone to errors and makes it difficult to tack available spaces and generate invoice efficiently. To address these challenges, the SmartPark needs a web based application that handles the parking management process. The system should allow the parking sales manager to record car entries and exits digitally, automatically calculate parking fees and update slots availability in real time.

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Task: Develop Parking Space Sales Management System (PSSMS)

As a full stack developer, you are given 7 hours to develop that web based application by:

1. Using attributes provided below, design an Entity Relationship Diagram (ERD) that represents the relationship between their entities.
   - Identify appropriate **primary keys** and **foreign keys** based on the relationships among the entities.
   - Entities and attributes are:

     1. **ParkingSlot**(SlotNumber,SlotStatus)
     2. **Car**(PlateNumber, DriverName, PhoneNumber)
     3. **ParkingRecord**(EntryTime, ExitTime, Duration)
     4. **Payment**(AmountPaid, PaymentDate)

   - ERD should be drawn before using the computer.
   - ERD should be drawn on plain paper using pencils.

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- ERD should indicate cardinalities, relationships with correct symbols

2. Creating database called **PSSMS** with ParkingSlot, Car, PakingRecord, and Payment tables as designed in ERD.
3. Saving your work in your real names in a folder called **(FirstName_LastName_National_Practical_Exam_2025).**
4. Preparing React.js Front-end application development environment by installing required modules and dependencies
5. Preparing JavaScript runtime environment for Node.js
6. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.

- Menu bar/Pages of web application should include Car, ParkingSlot, ParkingRecord, Payment, Reports and Logout options.
- The web application should be responsive

7. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.
8. Developing backend and frontend of parking sales management system.
   - Name backend project folder as **backend-project** and frontend project folder as **frontend-project**.
   - Use Tailwind CSS to implement UI design.
   - Your backend application should communicate to MySQL database/MongoDB to perform CRUD operations of using Node.js runtime environment and express.js framework.

   Note: 1.Insert operation should be used on all four (4) forms (Car, ParkingSlot, ParkingRecord and Payment)
   2. Delete, update, retrieve operations should only be used on ParkingRecord form
9. Creating react.js components with UI features that will enable user to input his data in the tables above and display required reports.

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10. Creating a session based login user account having username and password.

- Password must be strong and encrypted

11. Integrating your react.js application to backend application using Axios to enable interaction of your react.js application with node.js application.
12. Generating a bill containing PlateNumber, EntryTime, ExitTime, Duration, AmountPaid and PaymentDate.

- Parking fee calculations must be hourly based.
- The parking fee is 500RWF per hour, and any duration under one hour is still charged the full hourly rate.

13. Generating daily parking payment report indicating PlateNumber, EntryTime, ExitTime, Duration and AmountPaid .

14. Removing permanently your project with its related configurations after being marked.

- Ask permission to the assessor before removing your project.

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Note to the assessor:

- An assessor may promptly intervene to prevent the unnecessary use of internet and any other injury/incident that may occur during assessment
- The assessor should allow lunch time (one hour) break for candidate
- The assessor should ensure that each candidate is given time to exhibit his/her project by answering the following question:
    1. Describe your product in brief.
    2. Which difficulties have you met along the implementation of the project
    3. How did you overcome the difficulties
- The assessor should make sure that the distance between candidates prevents cheating
- The assessor should collect all ERD drawn by the candidates together with the exported project as evidences.

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2. CONSUMABLE MATERIALS, TOOLS AND EQUIPMENT

2.1. Consumable materials

2.1. A. Reusable materials

SM	Item	Technical specifications	Unit	Quantity	Unit price
1					
2					
3					

2.1. B. Non Reusable materials

SN	Item	Technical specifications	Unit	Quantity	Unit price
1	Internet	Minimum downloading speed of 25Mbps	mbps	1GB/2 Candidates	920
2	Flash disk	16 GB	Piece	1/20	475

2.2. Tools

2.2. A. First Choice (ideal) Tools

S/N	Item	Technical specifications	Measuring Unit	Quantity
1	VS Code	Latest Version	NA	1
2	Node	Stable Version	NA	1
3	Postman	Latest version	NA	1

2.3. B. Alternative Tools

SN	Item	Technical specifications	Measuring Unit	Quantity	Similar tools
1	Sublime text editor	Latest Version	NA	1	Any Text editor
2	Node	Stable version	NA	1	
3	Thunder client	Extension in vs code	NA	1	

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2.3. Equipment

2.3. A. First Choice (ideal) equipment

SN	Item	Technical specifications	Measuring unit	quantity
1	Laptop or Desktop Computer	i5 and above RAM: 4GB HDD :500GB		1
2	Generator	Automatic		1

2.3.B. Alternatives equipment

SN	Item	Technical specifications	Measuring unit	Quantity	Similar Equipment
1	Laptop or Desktop Computer	Intel,I3 RAM: 4GB HDD :500GB		1	
2	Generator	Manual		1	
3	UPS	8 hours of saving		1	

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ASSESSMENT CHECKLIST

Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
1.Preliminary activities performance (15%)						
	1.1.ERD is properly designed based on system requirements	Car entity is drawn			**1**	
		ParkingSlot entity is drawn			**1**	
		ParkingRecord entity is drawn			**1**	
		Payment entity is drawn			**1**	
		Entity Symbol is used			**1**	
		Relationship symbol is used			**1**	
		Link Symbol is used			**1**	
		Primary Key rule is respected			**1**	
		Foreign Key rule is respected			**1**	
		Cardinalities are indicated			**1**	
		Car and ParkingRecord relationship is indicated			**1**	
		ParkingSlot and ParkingRecord relationship is indicated			**1**	
		ParkingRecord and Payment relationship is indicated			**1**	
		ParkingSlot primary key is indicated			**1**	
		Car primary key is indicated			**1**	
		Payment primary key is indicated			**1**	
		ParkingRecord primary key is indicated			**1**	
		Car foreign key is indicated in ParkingRecord			**1**	

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Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		ParkingRecord foreign key is indicated in payment			1	
		ParkingSlot foreign key is indicated in ParkingRecord			1	
Subtotal					**.../20**	
Percentage (15%)					**.....**	
2. Process and fulfillment of the task (50%)						
	2.1.Development environment is properly arranged based on coding architecture methodology	Project folder is named as FirstName_LastName_National_Practical_Exam_2025			1	
		Node.js project is created			2	
		Express.js is installed			1	
		Cors is installed			1	
		Nodemon is installed			1	
		MySQL/MongoDB is installed in Node.js			1	
	2.2.React.JS environment is properly prepared based on application requirements.	React project is created			2	
		React-router-dom is installed			1	
		Axios is installed			1	
	2.3.Data definition language commands are effectively applied based on database schema	PSSMS database is created			1	
		User table is created			1	
		Car table is created			1	
		Payment table is created			1	
		ParkingSlot table is created			1	
		ParkingRecord table is created			1	
		Primary key is applied in User table			1	
		Primary key is applied in Car table			1	
		Primary key is applied in ParkingSlot table			1	

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Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		Primary key is applied in Payment table			1	
		Primary key is applied in ParkingRecord table			1	
		ParkinRecord Foreign key is applied in Payment table			1	
		Car Foreign key is applied in ParkingRecord table			1	
		ParkingSlot Foreign key is applied in ParkingRecord table			1	
	2.4.React basics are correctly applied based on application requirements.	Function is declared			2	
		Return method is included			1	
		Function is exported			1	
		Component is mounted to the DOM			2	
		JSX is used inside return method			1	
		JSX is well structured			1	
		Login form is created			1	
		Car form is created			1	
		ParkingSlot form is created			1	
		ParkingRecord form is created			1	
		Payment form is created			1	
	2.5.UI navigation is correctly applied based on application requirements	Routes are configured			2	
		Routing is performed			1	
		Links are created			2	
		Navigation between components is enabled			2	
	2.6.Tailwind utility classes are correctly Applied based on UI Design.	Tailwind CSS is installed			1	
		Tailwind CSS is configured			1	
		Tailwind is imported in the project			1	
		Class elements are used in JSX			2	

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Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		Flexbox is applied			1	
	2.7.Responsive design principles are properly applied according to the required design.	Flexible grid layouts are used			1	
		Media breakpoints are removed			1	
		Readability is Considered			1	
		Interactive elements are applied			1	
	2.8.Tailwind styles for a unique look are accurately customized based on tailwind configuration	Attractive colors are used			1	
		Text formatting is performed			1	
		UI components are attractive			1	
		Hover state is used			1	
		Focus state is used			1	
	2.9.Server and database connections are properly established according to development environment	Server JS file is created			2	
		Express.js package is imported			1	
		Cors package is imported			1	
		Port number is identified			1	
		Express object is created			1	
		Listen method is applied			1	
		MySQL/MongoDB package is used			2	
		Create connection method is used			1	
		Host parameter is passed			1	
		Username parameter is passed			1	
		Password parameter is passed			1	
		Database parameter is passed			1	
		Messages are displayed			1	
	2.10.RESTFUL APIs are effectively implemented based on backend functionalities	POST endpoint is created			2	
		GET endpoint is created			2	
		PUT endpoint is created			2	
		DELETE endpoint is			2	

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Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		created				
		Insert statement is used			2	
		Select statement is used			2	
		Update statement is used			2	
		Delete statement is used			2	
	2.11.API integration is properly implemented based on user requirements	API requests folder is created			2	
		User API JS file is created			1	
		Car API file is created			1	
		ParkingSlot API file is created			1	
		Payment API file is created			1	
		ParkingRecord API file is created			1	
		Axios is imported			1	
		API client is set			1	
		POST http method is used			2	
		Base URL parameter is passed			2	
	2.12.User Authentication, Authorization and Accountability (AAA) are carefully applied based on NPM Universal Access Control (UAC)	Form data is validated			2	
		Password input is encrypted			1	
		Duplicate data entry is eliminated			2	
		Password recovery is implemented			2	
Subtotal					**.../111**	
Percentage (50%)					**.....**	
3. Product presentation/Exhibition and Quality assessment (30%)						
	3.1. Project is effectively presented	The product name is mentioned			1	
		The key steps of the process are stated			3	
		The use/function/application			2	

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Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		/importance of the product is stated				
		Challenges/difficulties met are stated			1	
		Ways to overcome the challenges are stated			1	
		Candidate voice is audible			1	
		Body language is used			1	
	3.2. Usability is correctly tested according to expected results	Input validation prevents invalid data			2	
		Form submission saves data			2	
		Data retrieval displays records			2	
		Update functionality works			2	
		Delete functionality works			2	
	2.4. UI navigation is correctly applied based on application requirements	Navigation menus are interactive			1	
		Buttons are interactive			1	
		Fonts style is consistent			1	
		Intuitive colors are used			1	
		UI colors are consistent			1	
		Bill is generated to the driver			3	
		Generated bill is relevant			3	
		Daily based parking sales report is generated			3	
		Generated daily based parking sales report is relevant			3	
Sub Total					**.../37**	
Percentage (30%)					**.....**	
4.Closing activities (5%)						
	4.1.Project configurations and its dependencies are properly removed	Project folder is removed			1	
		Database is deleted			1	
		Global dependencies are removed			1	
		Text editor is not			1	

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Assessment criteria	Indicators	Elements of indicators	Observation:		Marks Allocation:	
			Yes	No	Max	Scored
		uninstalled				
		XAMPP is not uninstalled			1	
	4.2.Workplace laws and Time use are properly respected	Paper is free of excessive erasures			1	
		Waste materials are disposed off			1	
		Unnecessary creases or tears are avoided on papers used.			1	
	4.3.Time is properly managed	Time is respected			2	
Subtotal					**.../11**	
Closing Activities (5%)					**.....**	
Total Marks					**100**	
Percentage						

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4. PRE-DETERMINED CONDITIONS

General Environmental factors	Specific Equipment	Specific Tools
(1) Adequate Workshops and Laboratory space	-One desktop or laptop computers/student -Standby desktop or laptop computers for interventions in case another fails	Web browser, MySQL/MongoDB ,VS code/Sublime text editor
(2) Power source (Electricity + Generator)	Generator and UPS for those who use desktop computers	Automatic Generator and UPS of 1 hour power saving
(3) Availability of reliable internet and technological resources	Wired or Wireless internet	Cables or Modem
(4) Safety measures and protocol in place	The workspace should be designed to prevent liquids from coming into contact with electronic equipment, ensuring safety and longevity.	NA
(5) Office facilities (Computers, Printers, photocopying machines, Stapling machines)	Computers, Printers, photocopying machines, Stapling machines	Computer
(6) External storage device	External hard disk driver	NA

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